



Taiwanese Bankruptcy Prediction

The data were collected from the Taiwan Economic Journal for the years 1999 to 2009. Company bankruptcy was defined based on the business regulations of the Taiwan Stock Exchange.

Binary Classification Problem



Our problem is binary classification, where we predict if a company will go bankrupt or not based on the data

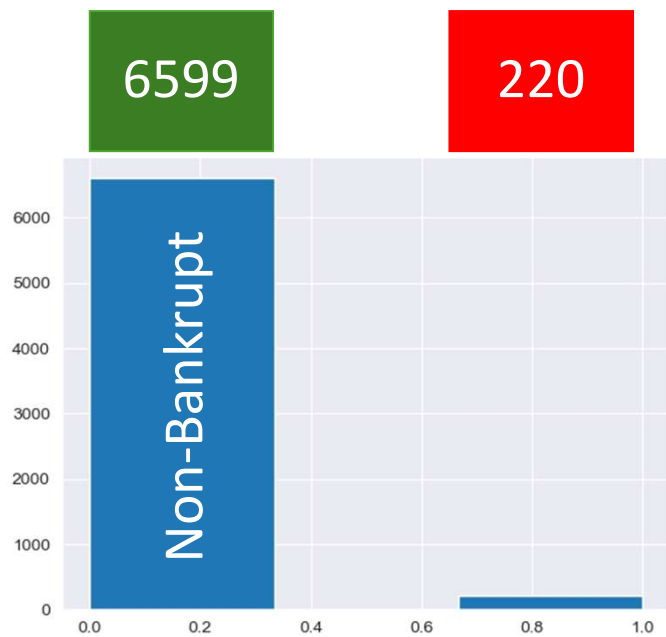
Instances**6819****Features****95****Missing Values****NO****Redundant Features****YES****X94 - Total Equity Growth = 1 in every row****X95 - Liability-Assets Flag = 0 in almost every row****Features values disproportion****Column with the highest mean value: X29 – Equity/Total Assets****Mean: 5508096595.24****Column with the lowest mean value: X85 – Cash Flow to Liability****Mean: 0.0011731925**

We have 6,819 instances and 95 features, with no missing values.

However, two of the columns are unnecessary

Additionally, there is a significant imbalance in the distribution of values across the columns.

Disproportion
between classes



Our dataset is also **imbalanced in terms** of the distribution of the negative and positive classes.

Among the all companies 6,599 have gone bankrupt, while 220 have not.

This means that only a small percentage of companies **3.23% had bankrupt**

Challenges of Evaluating Imbalanced Classes in Binary Classification

	precision	recall	f1-Score	support
0				
1				
accuracy				
macro avg				
weighted avg				

There are some challenges with evaluating imbalanced dataset.

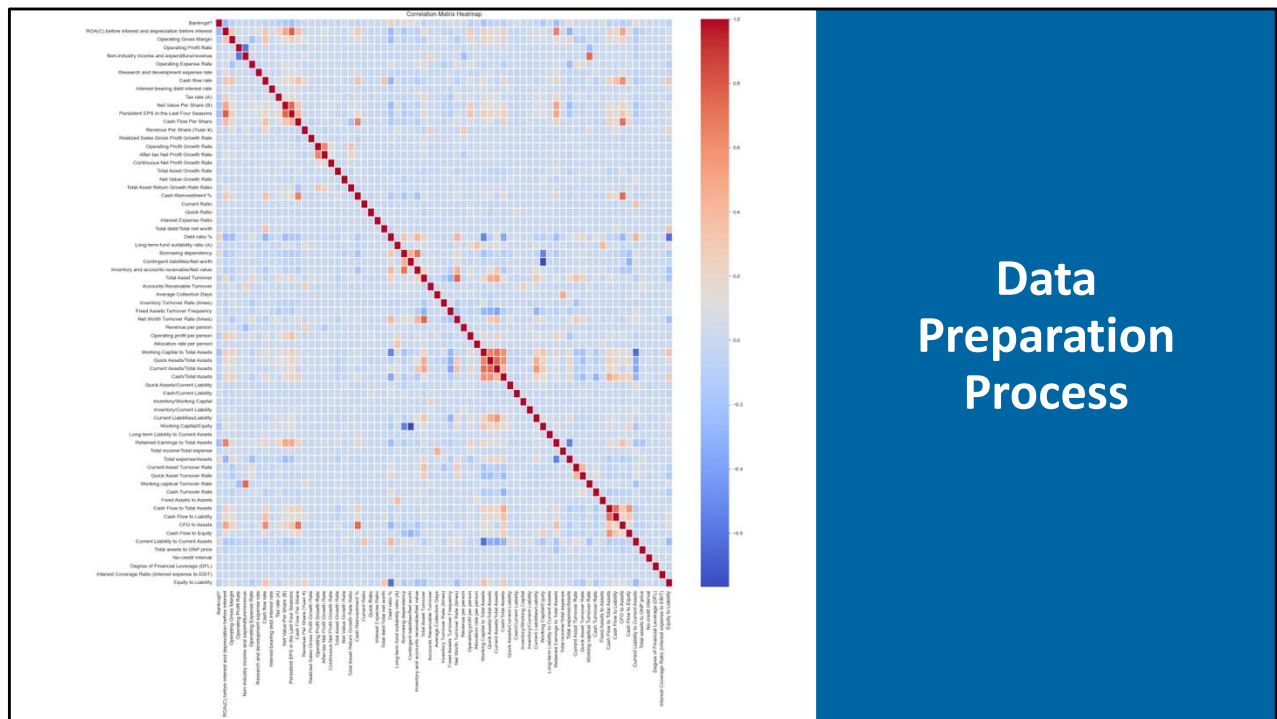
When classes are imbalanced , accuracy isn't a good metric because it can be misleading. In such cases, it's better to focus on precision and recall for each class individually **BECAUSE Precision** tells us how many of the things the model said were positive are actually positive. **AND Recall** tells us how many of the real positive cases the model found.

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First Evaluation on Raw Data Algorithm : LightGBM

	precision	recall	f1-Score	support
0	0.97	0.99	0.98	1313
1	0.62	0.29	0.40	51
accuracy			0.97	1364
macro avg	0.80	0.64	0.69	1364
weighted avg	0.96	0.97	0.96	1364

Here we can see the result of evaluation on Raw Data
The accuracy is 97%, but it's clear that it's not a valid indicator because **the model has trouble** identifying companies that went bankrupt AND model tends to favor the majority class due to imbalance. **Our task in** this project was to make this imbalance as insignificant as possible.



We removed columns with a linear correlation higher than 80% we considered it's too high based on the correlation histogram. **We tested** different correlation thresholds, **but we have to admit** that this step wasn't crucial and didn't significantly affect the results. It mainly helped reduce the data size, speeding up computations.

```
scaler =  
StandardScaler()
```

For each column
independently

mean = 0
standard deviation = 1

We used Standard Scaler for **each column independently** to standardize the data. **We had achieved** an overall improvement in the model's performance. **We also experimented** with different standardization methods for some groups of features but in finale case our best models use only StandardScaler().

Things worth mentioning that led us to the best results

- The use of the SMOTE method to generate synthetic examples for the minority class, which improves the data balance.
- The appropriate selection of hyperparameters means a lot
- In this case, removing outliers, which were numerous, negatively affected the recall as it resulted in a significant reduction in the amount of data available for training.
- During model training, we used the average precision metric, which combines precision and recall (PR AUC), meaning the model optimized average precision as its evaluation function to achieve a better performance assessment in the case of imbalanced data.

Removing many outliers reduced the data for training and lowered recall.

Last Evaluation on Edited Data with fine tuned parameters, Algoritm: LightGBM

	precision	recall	f1-Score	support
0	0.99	0.99	0.99	662
1	0.62	0.75	0.68	20
accuracy			0.98	0.98
macro avg	0.80	0.87	0.83	682
weighted avg	0.98	0.98	0.98	682

Model:
LightGBM

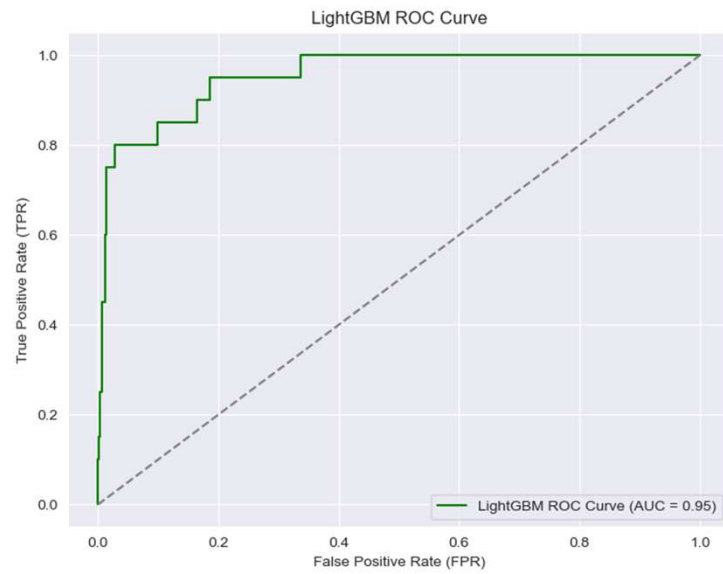
Corr:
> 80 removed

Sets:
80% Train
10% Val
10% Tests

Synthetic
Sampling:
Smote

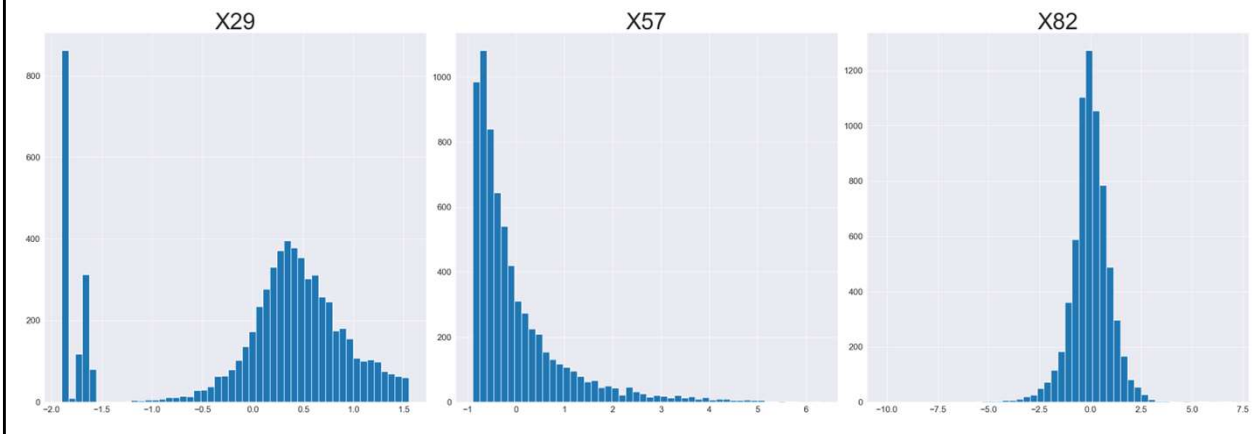
DANE podział val -validation

Our final version after editing the data for LightGBM: we managed to get the most balanced results, which are well-suited for our dataset. Additionally, the model shows good overall performance across various metrics, indicating its robustness and reliability for this particular task. This balance in the metrics suggests that the model is generalizing well and can be trusted to make accurate predictions in real-world scenarios.



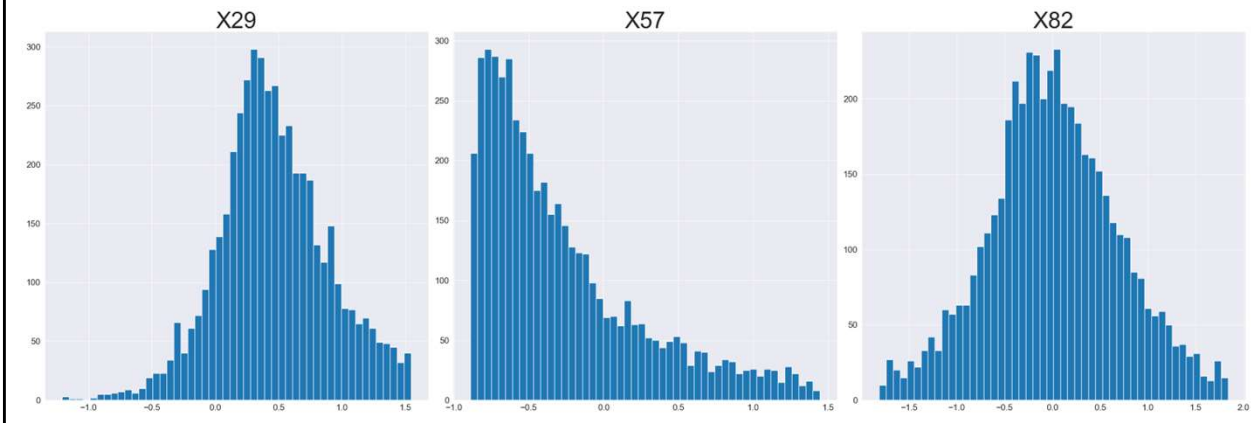
AUC=0.95 means the model works well and clearly separates the classes, not guessing randomly.

Impact of Outlier Removal



We selected one more model based on the previous one, but this time we removed outliers from the selected columns

Rows before removing outliers: 6819
Rows after removing outliers: 4850



We lost about 30% of the samples, but it increase precision in detecting companies but decrease Recall.

Evaluation on Edited Data with fine tuned parameters with Outliners Removal

	precision	recall	f1-Score	support
0	0.98	0.99	0.99	467
1	0.77	0.56	0.65	18
accuracy			0.98	0.98
macro avg	0.87	0.77	0.81	485
weighted avg	0.98	0.98	0.98	485

Model:
LightGBM

Corr:
> 80 removed

Sets:
80% Train
10% Val
10% Tests

**Synthetic
Sampling:**
Smote

The model correctly classifies only 56% of all actual bankrupt companies, but if it identifies one as bankrupt, it is correct 77% of the time.

Conclusions:

1. If the goal is to identify as many companies that are likely to go bankrupt as possible (minimizing false negatives), a model with higher **recall** will be better. It may include more false positives, but it reduces the chance of missing bankruptcies.
2. If the priority is to ensure that every bankruptcy prediction is highly reliable (minimizing false positives), a model with higher **precision** is preferred. This may miss some bankruptcies but ensures that predicted bankruptcies are more likely to be accurate

Wnioski :

Gdy zależy nam aby spośród wszystkich firm znaleźć więcej tych, które zbankrutują prawdopodobnie lepszym modelem będzie ten, który ma większy RECALL

- To find more bankrupt companies, choose a model with high recall. It may include more false positives, but it reduces the chance of missing bankruptcies.
- To be sure predicted bankruptcies are correct, choose a model with high precision. It may miss some cases but will be more accurate.

Jeżeli bardziej zależy nam aby każda predykcja była wysoko prawdopodobna ale wykrywalność firm, które zbankrutują niższa to model z większą precyzją